3.3. GLYCOLYSIS

Glycolysis is an oxidative metabolic process in which one mole of glucose is converted into two moles of pyruvate with the generation of chemical energy in the form of ATP in a series of ten enzyme catalysed reactions. In this process the oxidising agent is coenzyme NAD^+. No molecular oxygen is utilised in glycolysis and hence it is an anaerobic pathway.

Under aerobic conditions, the pyruvate formed by glycolysis is further oxidized to CO_2 in the citric acid cycle. Whereas, in anaerobic conditions, the pyruvate is converted to lactate. This process occurs in the cytoplasm of all the cells.

3.3.1. Reactions of glycolysis pathway

There are two stages in the glycolysis namely preparatory stage and pay off stage.

Preparatory Stage:

It consists of first five reactions in which one molecule of glucose is converted into two molecules of glyceraldehyde-3-phosphate. This stage is an energy consuming stage in which two ATP molecules are utilized.

Pay off Stage:

It consists of the remaining five reactions in which the glyceraldehyde-3-phosphoate is converted into pyruvate. This is an energy producing stage in which two molecules of ATP are produced from ADP, per glyceraldehyde-3-phosphate. i.e. for two glyceraldehyde-3-phosphate molecule, totally four ATP molecules are produced.

In the overall glycolysis process there is a net gain of two ATP molecules for conversion of every glucose molecules into pyruvate. The overall process can be represented by the following equation.

$$\text{Glucose} + 2NAD^+ + 2ADP + 2P_i \rightarrow 2 \text{ pyruvate} + 2NADH + 2\ ATP + 2H^+ + 2H_2O$$

Reactions of Preparatory stage:

Step 1: Glucose is phosphorylated by the transfer of a phosphate group from ATP molecule to form glucose-6-phosphate and ADP. This step is energy consuming step and the required energy is provided by the breakdown of the ATP molecule. Glucose-6-phosphate is more reactive than glucose and also cannot cross the cell membrane. Hence it is trapped inside the cell. This step is catalysed by hexakinase enzyme.

Step 2: Glucose-6-phosphate is isomerised to form fructose-6-phosphate. This step is catalysed by the enzyme phosphogluco isomerase

Step 3: Like the first step the fructose-6-phosphate is phosphorylated again by ATP to form fructose-1,6-bisphosphate. This step is catalyzed by the enzyme phosphofructokinase.

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